MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST (SET – A)** | **Time:** 1 Hour
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. This question paper contains 13 compulsory questions divided into three sections: A, B, and C.
2. **Section A** comprises 5 questions of 1 mark each (Q1–Q3 are MCQs; Q4–Q5 are Assertion–Reasoning).
3. **Section B** comprises 5 questions of 2 marks each (Short Answer Type questions).
4. **Section C** comprises 3 questions of 5 marks each (Long Answer Type questions).
5. Use of calculators or electronic devices is strictly prohibited. Draw neat figures wherever required.

SECTION A: 1 Mark Questions (3 MCQs + 2 A–R) (5 × 1 = 5 Marks)

Q1. The distance between the points $(0, 5)$ and $(-5, 0)$ is: [1]

(A) 5 (B) $5\sqrt{2}$ (C) $2\sqrt{5}$ (D) 10

Q2. The coordinates of the point which divides the line segment joining the points $(1, 3)$ and $(4, 6)$ internally in the ratio $2 : 1$ are: [1]

(A) $(2, 4)$ (B) $(3, 5)$ (C) $(4, 2)$ (D) $(5, 3)$

Q3. If the midpoint of the line segment joining the points $A(3, 4)$ and $B(k, 6)$ is $P(x, y)$ and $x + y - 10 = 0$, then the value of k is: [1]

(A) 6 (B) 7 (C) 8 (D) 9

Directions for Q4 & Q5: A statement of Assertion (A) is followed by Reason (R). Choose:
(A) Both (A) and (R) are true and (R) is correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not correct explanation of (A).
(C) (A) is true but (R) is false. **(D)** (A) is false but (R) is true.

Q4. **Assertion (A):** The distance between the points $P(-2, 0)$ and $Q(2, 0)$ is 4 units.
Reason (R): Mid-point of a line segment divides it in the ratio $1 : 1$. [1]

Q5. **Assertion (A):** The point $(0, 2)$ is the point of intersection of the y-axis with the line segment joining $(1, 1)$ and $(-1, 3)$.
Reason (R): The coordinates of a point dividing the line segment joining (x_1, y_1) and (x_2, y_2) in the ratio $m_1 : m_2$ is given by $\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$. [1]

SECTION B: Short Answer Type Questions (5 × 2 = 10 Marks)

Q6. Find the distance between the points $P(a, b)$ and $Q(-a, -b)$. [2]

Q7. If the point $A(0, 2)$ is equidistant from the points $B(3, p)$ and $C(p, 5)$, find the value of p. Also find the length of line segment AB. [2]

SECTION B (Continued): Short Answer Type Questions (Questions 8 to 10)

Q8. Find the ratio in which the y-axis divides the line segment joining the points $A(5, -6)$ and $B(-1, -4)$. Also find the coordinates of the point of division. **[2]**

Q9. Find the coordinates of a point A, where AB is the diameter of a circle whose centre is $(2, -3)$ and B is $(1, 4)$. **[2]**

Q10. If the midpoint of the line segment joining $A(2a, 4)$ and $B(-2, 3b)$ is $M(1, 2a + 1)$, find the numerical values of a and b. **[2]**

SECTION C: Long Answer Type Questions ($3 \times 5 = 15$ Marks)

Q11. If the point $P(x, y)$ is equidistant from the points $A(a + b, b - a)$ and $B(a - b, a + b)$, prove rigorously that:

$$bx = ay$$

[5]

Q12. Answer the following questions regarding points of trisection:

(i) Define what is meant by the points of trisection of a line segment. **[1]**

(ii) Find the coordinates of the points of trisection of the line segment joining the points $A(2, -2)$ and $B(-7, 4)$. **[4]**

Q13. The vertices of a rhombus taken in order are $A(3, 0)$, $B(4, 5)$, $C(-1, 4)$ and $D(-2, -1)$.

(i) Calculate the lengths of its diagonals AC and BD using the distance formula. **[3]**

(ii) Hence, find the area of the rhombus. $\left[\text{Hint: Area of rhombus} = \frac{1}{2} \times d_1 \times d_2\right]$ **[2]**

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